Nutrition Assistant Project
Phase 6: Performance & Testing

Introduction

The Performance and Testing phase is conducted to verify that the Nutrition Assistant application functions correctly, performs efficiently, and satisfies all functional and non-functional requirements. Quality Assurance (QA) activities help identify defects, improve system reliability, and ensure a smooth user experience before deployment.

Objectives

Verify application functionality. Identify and fix software defects. Ensure system performance under different workloads. Validate security and data integrity. Improve usability and accessibility.

Testing Strategies

Unit Testing: Individual components, utility functions, and business logic are tested using frameworks such as Jest to ensure each module behaves correctly in isolation.

Integration Testing: Communication between frontend, backend, and database is verified to ensure all modules work together seamlessly.

API Testing: REST API endpoints are tested using Postman or Thunder Client to validate request handling, authentication, response codes, and error management.

Load Testing: Multiple concurrent users and requests are simulated to evaluate application scalability and response time under heavy traffic.

User Acceptance Testing (UAT): End users validate the application to confirm it satisfies business requirements and delivers the expected functionality.

Performance Evaluation

Application performance is measured by monitoring API response times, database query execution, frontend rendering speed, and resource utilization. Optimization techniques such as caching, lazy loading, image compression, and efficient database indexing are implemented to improve overall performance.

Lighthouse Evaluation

The frontend application is evaluated using Google Lighthouse with the following target scores: **Performance:** 90+ for fast loading and efficient rendering. **Accessibility:** 90+ by following accessibility standards. **Best Practices:** 90+ through secure and modern web development. **SEO:** 90+ by implementing search engine optimization guidelines.

Bug Tracking and Quality Assurance

Detected defects are recorded, prioritized, and resolved using issue tracking tools such as GitHub Issues or Jira. Regression testing is performed after bug fixes to ensure new changes do not introduce additional defects.

Conclusion

The Performance and Testing phase ensures that the Nutrition Assistant application is reliable, secure, scalable, and user-friendly. Comprehensive testing improves software quality, reduces production issues, and increases user satisfaction by delivering a stable and high-performing application.